6.S974 Games, Learning, and Security
HW2

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Problem 1 (Cybersecurity with Incomplete Information)

(a) **Pure strategies.** A strategy specifies a choice for each type.

- Attacker ($\theta_1 = 4, \theta_1 = 8$): $\{(A, A), (A, N), (N, A), (N, N)\}$
- Company: ($\theta_2 = 1, \theta_2 = 3$): $\{(P, P), (P, U), (U, P), (U, U)\}$

A strategy profile is a tuple of one pure strategy per player. In this case each player has four pure strategies, so there are $4 \times 4 = 16$ strategy profiles.

(b) **Prior and beliefs.** Types are equally likely to occur and independent so

$$p(\theta_1, \theta_2) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4}, \qquad \forall\, \theta_1,\ \forall\, \theta_2$$

Beliefs are conditional on the attacker or the company's types

$$\mu_1(\theta_2 = 1 \mid \theta_1 = 4) = \frac{p(4,1)}{p(4,1) + p(4,3)} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \tfrac{1}{2}$$

Because the prior is uniform over (θ_1, θ_2), this belief holds for every player's type for every other player's type so

$$\mu_1(\theta_2 \mid \theta_1) = \mu_2(\theta_1 \mid \theta_2) = \tfrac{1}{2}$$

(c) **Pure strategy best response.** Expected payoff when company plays P is $E_{A\sim\alpha}[u_2(P, A)] = -\alpha\theta_2 - (1-\alpha)\theta_2 = -\theta_2$ and when company plays U is $E_{A\sim\alpha}[u_2(U, A)] = -4\alpha$, so

$$BR_2(\alpha) = \begin{cases} P & \theta_2 < 4\alpha \\ \{P, U\} & \theta_2 = 4\alpha \\ U & \text{else} \end{cases} \implies \begin{cases} (P,P) & \alpha \geq \frac{3}{4} \\ (P,U) & \frac{1}{4} \leq \alpha \leq \frac{3}{4} \\ (U,P) & \text{never} \\ (U,U) & \alpha \leq \frac{1}{4} \end{cases}$$

And for the attacker we get $E_{P\sim\pi}[u_1(P, A)] = -\pi + (1-\pi)(\theta_1 - 1) = (1-\pi)\theta_1 - 1$ and $E_{P\sim\pi}[u_1(P, N)] = 0$

$$BR_1(\pi) = \begin{cases} A & \pi < 1 - \frac{1}{\theta_1} \\ \{A, N\} & \pi = 1 - \frac{1}{\theta_1} \\ N & \text{else} \end{cases} \implies \begin{cases} (A,A) & \pi \leq \frac{3}{4} \\ (A,N) & \text{never} \\ (N,A) & \frac{3}{4} \leq \pi \leq \frac{7}{8} \\ (N,N) & \pi \geq \frac{7}{8} \end{cases}$$

(d) **Does a pure strategy Bayes Nash Equilibrium exist?**

- Begin with attacker pure strategy (A, A): $\sigma_1(A \mid \theta_1 = 4) = 1 = \sigma_1(A \mid \theta_1 = 8)$ so $\alpha = 1$. From part (c) $BR_2(1) = (P, P)$. By the same logic company pure strategy (P, P) yields $\pi = 1$ $BR_1(1) = (N, N)$. Repeating, attacker pure strategy (N, N) yields $\alpha = 0$ and $BR_2(0) = (U, U)$. Company pure strategy (U, U) yields $\pi = 0$ and $BR_1(0) = (A, A)$, returning us to the start of the cycle. So there's no pure strategy Bayes Nash Equilibrium with any combination of $((A, A), (N, N))$ and $((P, P), (U, U))$
- Take attacker pure strategy (A, N) yielding $\alpha = \frac{1}{2}$ and $BR_2(\frac{1}{2}) = (P, U)$. The company's best response yields $\pi = \frac{1}{2}$ and $BR_1(\frac{1}{2}) = (A, A)$, returning us to the first cycle.
- Take attacker strategy (N, A) yielding $\alpha = \frac{1}{2}$. This produces the same $BR_2(\frac{1}{2}) = (P, U)$ above returning us to the first cycle.

Hence by process of elimination, there is no pure strategy Bayes Nash Equilibrium.

(e) **Bayesian Nash Equilibrium in behavioral strategies.**

Necessary condition $\pi = \frac{3}{4}$ By contradiction suppose $\pi \neq \frac{3}{4}$ in equilibrium. Recall that a mixed action is a best response only if every action in its support is a pure best response by theorem.

Case 1 $(\pi > 3/4)$**.** For any $\pi > 3/4$ attacker type $\theta_1 = 4$ has only a pure best response N (because mixed best response requires pure best response in its support) and $\sigma_1(A \mid \theta_1 = 4) = 0$. So $\alpha = \sigma_1(A \mid \theta_1 = 8)/2$, which implies $\alpha \leq 1/2$.

For any $\alpha \leq 1/2$ the company type $\theta_2 = 3$ has only a pure best response U and $\sigma_2(P \mid \theta_2 = 3) = 0$. So $\pi = \sigma_2(P \mid \theta_2 = 1)/2$, which implies $\pi \leq 1/2$, a contradiction to $\pi > 3/4$.

Case 2 $(\pi < 3/4)$ For any $\pi < 3/4$ the attacker has only a pure best response (A, A) and $\alpha = 1$. For $\alpha = 1$ the company has only a pure best response (P, P) and $\pi = 1$, contradicting $\pi < 3/4$.

Therefore if there exists a Bayesian Nash Equilibrium a necessary condition would be $\pi = 3/4$.

Mixed strategy. The probability of protecting $\pi = 3/4$ gives $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) = 3/2$, which implies at least one company type randomizes, since pure company strategies give $\sigma_2(P \mid 1) + \sigma_2(P \mid 3) \in \{0, 1, 2\}$.

Moreover, for $\pi = 3/4$ the best response of attacker type $\theta_1 = 8$ is A, implying a probability of attack $\alpha \geq 1/2$.

For the company to have a mixed strategy and $\alpha \geq 1/2$, it must be that $\alpha = 3/4$ so that the mixed strategy best response can have a support greater than 1: the company mixes best responses $(P, P), (P, U)$.

This support tells us that company type $\theta_2 = 1$ has a pure strategy best response of P and $\sigma_2(P \mid 1) = 1$. Plugging this into our equation for π above we get $\sigma_2(P \mid 3) = 1/2$.

Note that $\sigma_1(A \mid 8) = 1$ follows directly from attacker type $\theta_1 = 8$'s best response of A. Solving $\alpha = 3/4 = (1 + \sigma_1(A \mid 4))/2$ gives $\sigma_1(A \mid 4) = 1/2$.

Therefore if there is a mixed strategy Bayesian Nash Equilibrium it must be that

$$\sigma_2(P \mid 3) = 1/2 = \sigma_1(A \mid 4), \quad \sigma_2(P \mid 1) = 1 = \sigma_1(A \mid 8).$$

Suffice to validate.

Validation. From definition, the strategy profile is a Bayes Nash Equilibrium if for every player and every type the strategy is a best response. At the candidate profile $\pi = 3/4$ and $\alpha = 3/4$.

$$\begin{aligned}
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 4)) = \{A, N\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 4) = \{A, N\} \\
\text{supp}(\sigma_1(\cdot \mid \theta_1 = 8)) = \{A\} &\subseteq BR_1(\tfrac{3}{4} \mid \theta_1 = 8) = \{A\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 1)) = \{P\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 1) = \{P\} \\
\text{supp}(\sigma_2(\cdot \mid \theta_2 = 3)) = \{P, U\} &\subseteq BR_2(\tfrac{3}{4} \mid \theta_2 = 3) = \{P, U\}
\end{aligned}$$

Each condition above was necessary for equilibrium, so it is the unique

Problem 2 (Signaling through Safety Certification)

(a) **Game tree.**

(b) **Does a separating PBE exist?**

(c) **Pooling PBE.**

(d) **Semi-separating equilibrium.**

Problem 3 (Traffic Routing Problem)

(a) **Waldrop equilibrium.**

(b) **Mean-posterior identity.**

(c) **Public signal rule.**